Visualizing Snowfall Prediction

Process Book

CS4804 Data Visualization, C26

Group Members: Karinne Aiello, Aditya Patel, Victor Radina

Overview and Motivation:

The goals of this project are to design an exploratory interface to increase users' understanding of snowfall predictions in their area, explore effective interactive visualizations and web development techniques, and support our project work with that of other experts in the field.

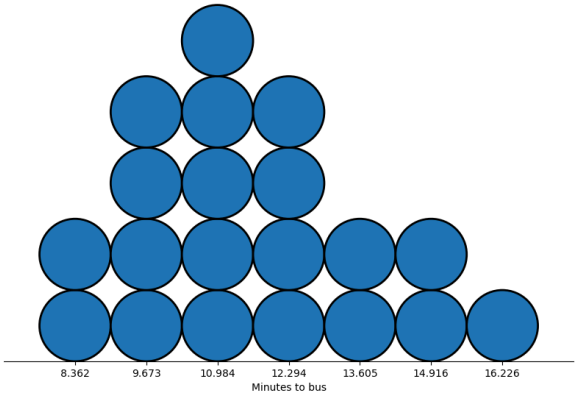
Our team initially wanted to visualize weather data, envisioning a map of America with different colors to represent temperature and precipitation levels and the interactivity necessary for users to switch between days. Given recent and upcoming snowstorms, along with many other weather maps available, we considered that an effective visualization for snowfall prediction could be more useful and original. One of our members checks the weather daily and has noticed that communication of snowfall prediction for these storms can be hard to understand. The professor introduced us to a newer and more interpretable method of data visualization called a quantile dot plot, which is the focus of our project. As this chart is unfamiliar to us, we expect minor challenges with implementation, but welcome the growth opportunity.

Related Work:

A 2016 research article by University of Washington and Tufts University professionals "When (ish) is My Bus?: User-centered Visualizations of Uncertainty in Everyday, Mobile Predictive Systems" proposed quantile dot plots as a novel discrete representation of continuous outcomes for small screens (Kay et al., 2016). Quantile dot plots visualize a probability distribution of a given outcome in what my group believes to be a more comprehensible medium for viewers. Through a literature review, survey of 172 users of realtime transit application OneBusAway, and an iterative design process, the authors identified important design requirements for visualizing uncertainty. The approach they adopted was similar to that of a 1987 study that queried subjects to report probabilities based on various representations of uncertainty for weather forecasts (Ibrekk & Morgan, 1987).

In their controlled experiment analyzing four methods of visualization–a dotplot with 20 instances, dotplot with 100 instances, density plot, and stripeplot–the data scientists found that respondents' estimates in the dotplot with 20 instances were the most precise of the four visualization types (Kay et al., 2016). The Dotplot-20 reduced the variance of probabilistic estimates by about 1.15 times compared to density plots, as well improved users' confidence in their estimations–with 81% confidence for the Dotplot-20 estimates and 73% for the next most confident figure, the Dotplot-100. With an 11% averaged standard deviation for the Dotplot-20, 13% for the Dotplot-100 and density plot, and 16% for the stripeplot, we can note that increasing the hypothetical instances visualized by a dotplot from 20 to 100 does not improve interpretability. As a result, our team will default our quantile dot plot to 20 dots, but we will allow users to adjust this amount if they would like.

The visuals below seek to answer the question "What time should I arrive at the bus stop in order to catch my bus?" which is the 2016 paper's objective. They illustrate a quantile dot plot of the probability a bus will take a certain number of minutes to arrive for 20 possible days, with each dot representing a hypothetical day.



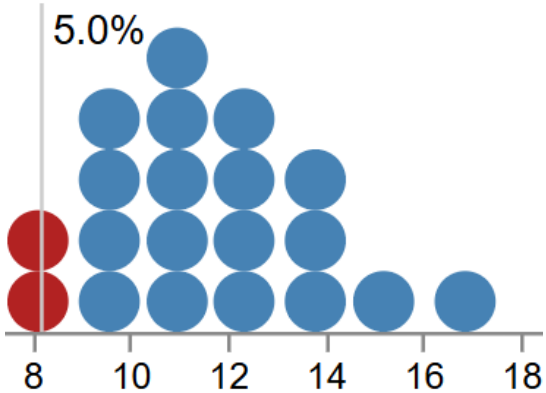


Figure 1: When Is My Bus? with detailed minute information (Carroll, 2019)
Figure 2: When Is My Bus? with 5% probability line (*Quantile Dot Plot Example*, n.d.)

Based on these figures, users can estimate a comfortable arrival time for themselves by selecting the level of risk they are okay with–such as missing the bus 10% of the time–and counting the outcomes to reach that value. If someone only wants to risk missing the bus 2 out of 20 times (10%), they can count up 2 dots/buses from the left to determine the time to arrive at their bus stop. In the case of snowfall, users could review the visualization to understand how many theoretical days out of a given amount will experience a certain level of precipitation. This information could better inform their decisions on days when snowfall is projected.

Questions:

The questions we originally wanted to answer include:

- Are quantile dot plots an effective visualization method for snowfall prediction?
- Do quantile dot plots help people better understand probabilistic snowfall forecasts compared to traditional bar charts?

Over the course of the project, these original questions evolved and we considered new inquiries, such as:

- Does a quantile dot plot being vertical as opposed to horizontal affect interpretability? Positively or negatively?

Data:

Our project uses user-inputted data for the likelihood of different snowfall levels to generate a quantile dot plot of the information. Websites and tools like the New York Times' "How Much Snow Will Fall Where You Live?" enable viewers to get this information in the form of bar charts (Bhatia et al., 2026). To find the amount of snow that truly ends up falling, the National Centers for Environmental Information - National Oceanic and Atmospheric Administration database provides free, nationwide climate data (National Centers for Environmental Information, n.d.).

Exploratory Data Analysis:

We used the New York Times tool to determine the likelihoods for snowfall levels in the range of February 20 to February 23 (Bhatia et al., 2026).

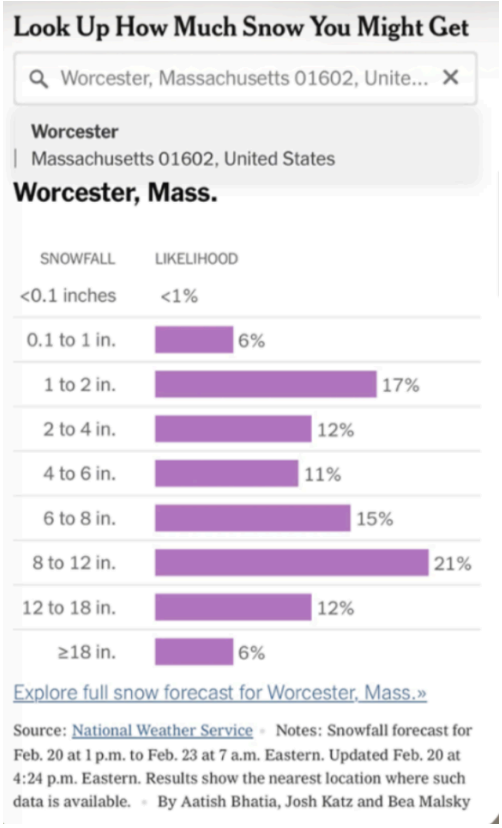


Figure 3: Snowfall Range Likelihoods for February 20-23, 2026 in Worcester, MA 01602 (Bhatia et al., 2026)

To determine what the actual snowfall was, we used the National Centers for Environmental Information - National Oceanic and Atmospheric Administration (NOAA)

database to request the climate data for February 20 to February 23–the range shown in the bar chart (National Centers for Environmental Information, n.d.).

U.S. Department of Commerce
National Oceanic & Atmospheric Administration
National Environmental Satellite, Data, and Information Service

Record of Climatological Observations
These data are quality controlled and may not be identical to the original observations.
Generated on 03/03/2026

National Centers for Environmental Information
151 Patton Avenue
Asheville, North Carolina 28801

Current Location: Elev: 1002 ft. Lat: 42.2706° N Lon: 71.8731° W
Station: **WORCESTER REGIONAL AIRPORT, MA US USW00094746**

Observation Time Temperature: Unknown Observation Time Precipitation: 2400

Year	Month	Day	Temperature (F) "24 Hrs. Ending at Observation Time" Max.	Temperature (F) "24 Hrs. Ending at Observation Time" Min.	Temperature (F) At Obs.	Precipitation 24 Hour Amounts Ending at Observation Time: Rain, Melted Snow, Etc. (in)	Precipitation: Flag	Precipitation: Snow, Ice Pellets, Hail (in)	Precipitation: Flag	Precipitation At Obs. Time: Snow, Ice Pellets, Hail, Ice on Ground (in)	Evaporation: 24 Hour Wind Movement (mi)	Evaporation: Amount of Evap. (in)	"Soil Temperature (F)" 4 in. Depth: Ground Cover (see *)	"Soil Temperature (F)" 4 in. Depth: Max.	"Soil Temperature (F)" 4 in. Depth: Min.	"Soil Temperature (F)" 8 in. Depth: Ground Cover (see *)	"Soil Temperature (F)" 8 in. Depth: Max.	"Soil Temperature (F)" 8 in. Depth: Min.
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2026	02	21	30	26		0.05		1.1										
2026	02	22	29	25		0.22		2.3										
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Empty, or blank, cells indicate that a data observation was not reported.
"Ground Cover: 1=Grass; 2=Fallow; 3=Bare Ground; 4=Brome grass; 5=Sod; 6=Straw mulch; 7=Grass muck; 8=Bare muck; 0=Unknown
"s" This data value failed one of NCEI's quality control tests. "At Obs." = Temperature at time of observation
"T" values in the Precipitation or Snow category above indicate a "trace" value was recorded.
"A" values in the Precipitation Flag or the Snow Flag column indicate a multiday total, accumulated since last measurement, is being used.
Data value inconsistency may be present due to rounding calculations during the conversion process from SI metric units to standard imperial units.

Figure 4: NOAA Climate Data for True Snowfall February 20-23, 2026 in Worcester, MA 01602 (National Centers for Environmental Information, n.d.)

Through our requested climate data, we determined the total amount of snow Worcester, Massachusetts 01602 actually received February 20-23, 2026 was 18.6 inches. Viewers of the New York Times bar chart would certainly not have predicted this level of snowfall, given only a 6% likelihood. The bar with the highest chance at 21% was eight to twelve inches, with the second at 17% being one to two inches, and the third at 15% being six to eight. As such, users' assumptions from the bar chart were likely positioned around nine inches–less than half of the true value.

We also examined the visualizations in our 2016 and 1987 related studies. The following image illustrates respondents' precision in estimating probabilities with quantile dot plots, along with density plots and stripeplots. As a result of the 2016 survey, the authors found that the quantile dot plot with 20 dots was easier for viewers to accurately interpret than the quantile dot plot with 100 dots, thus we will fix the number of dots for our visualization at 20 and not provide the option to change this, as initially suggested.

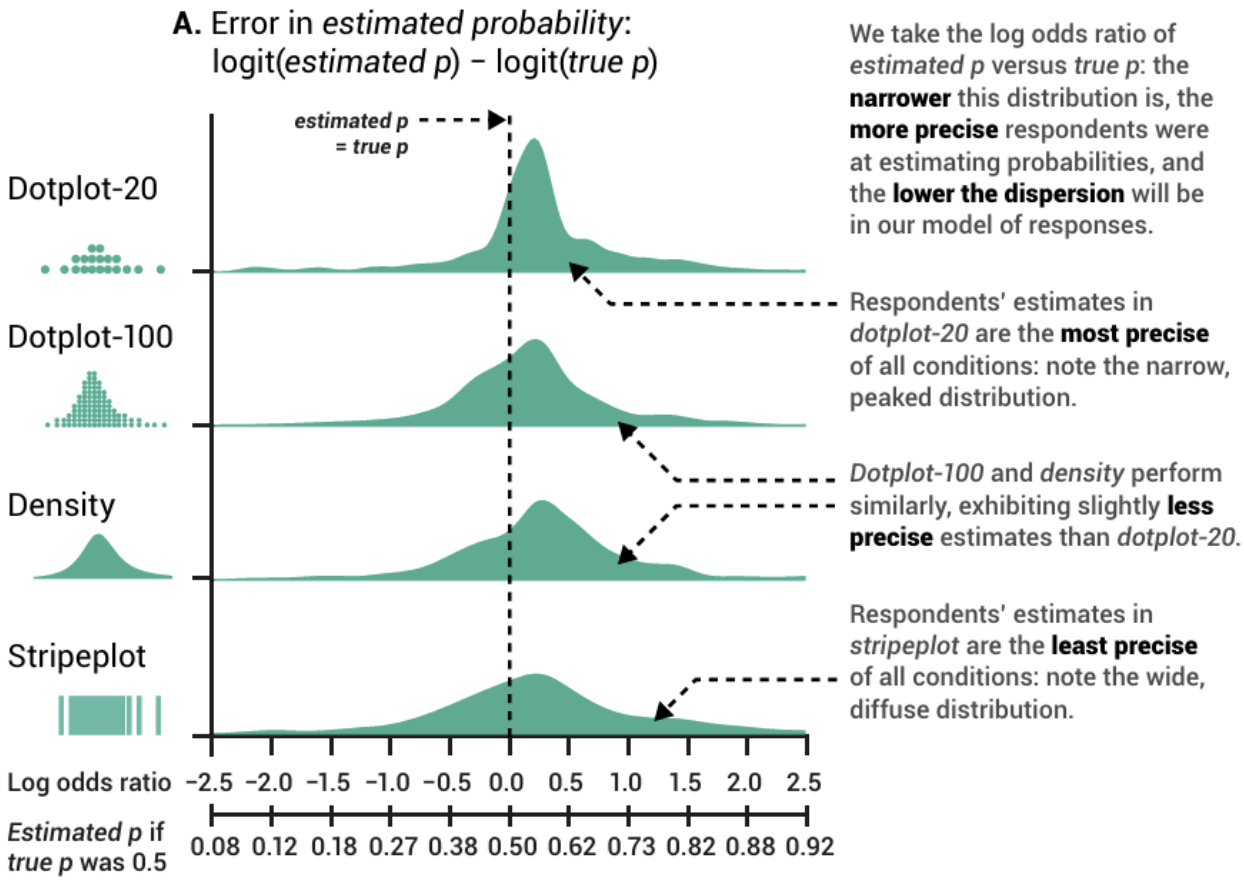

Figure 5: 2016 Study's Comparison of Quantile Dot Plots and Other Visualization Types (Kay et al., 2016)

Design Evolution:

The Design Activity Framework comprises understanding design requirements, ideating ideas, making prototypes, and deploying the visualization system (McKenna & Harrison, n.d.). This guides the process of problem-driven work for visualization design and supports a more flexible design process. We kept this framework in mind throughout our design evolution. To understand design requirements, we considered our unique needs and researched other papers investigating quantile dot plots and the accuracy of user estimations depending on the visualization type. While we originally expected to create a quantile dot plot with the dots aligned vertically, we ended up deviating and choosing to plot the dots horizontally for each snowfall level.

(1) We initially created a traditional, horizontal quantile dot plot to represent the 20 hypothetical days of snowfall. Users inputted values for the <0.1, 0.1-1, 1-2, 2-4, 4-6, 6-8, 8-12, 12-18, and >=18 levels by typing the respective likelihoods in each corresponding box. We also sketched out a framework for our final website, including the visualizations and additional requirements.

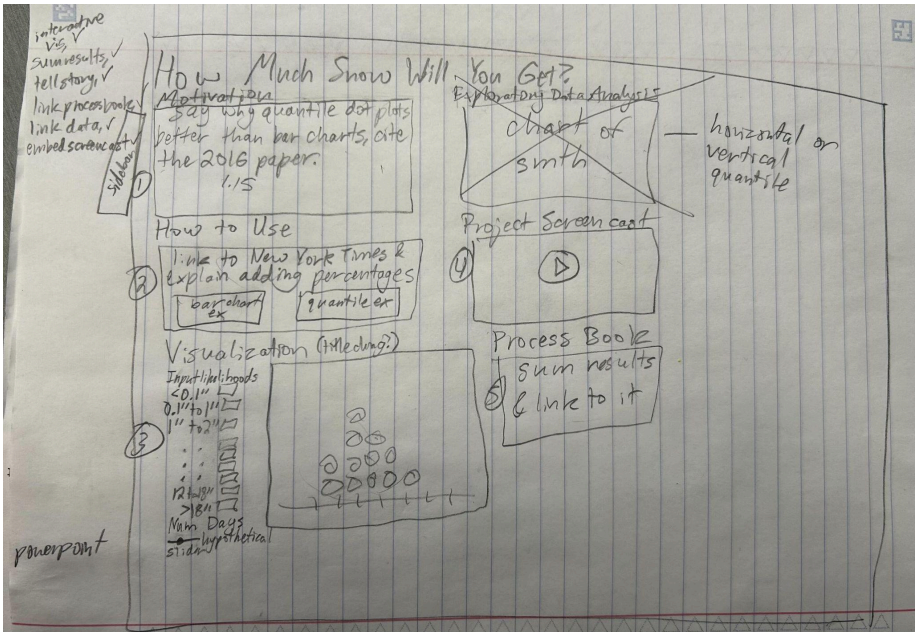

Figure 6: Detailed Website Design Sketch

(2) Given recommendations from the professor, design principles from class, and our own intuitions, however, we chose to create a vertical quantile dot plot next. This better draws users' eyes to the snowfall level with the greatest number of days and makes a better comparison between the quantile dot plot and the bar chart, as the bar chart is horizontal instead of vertical.

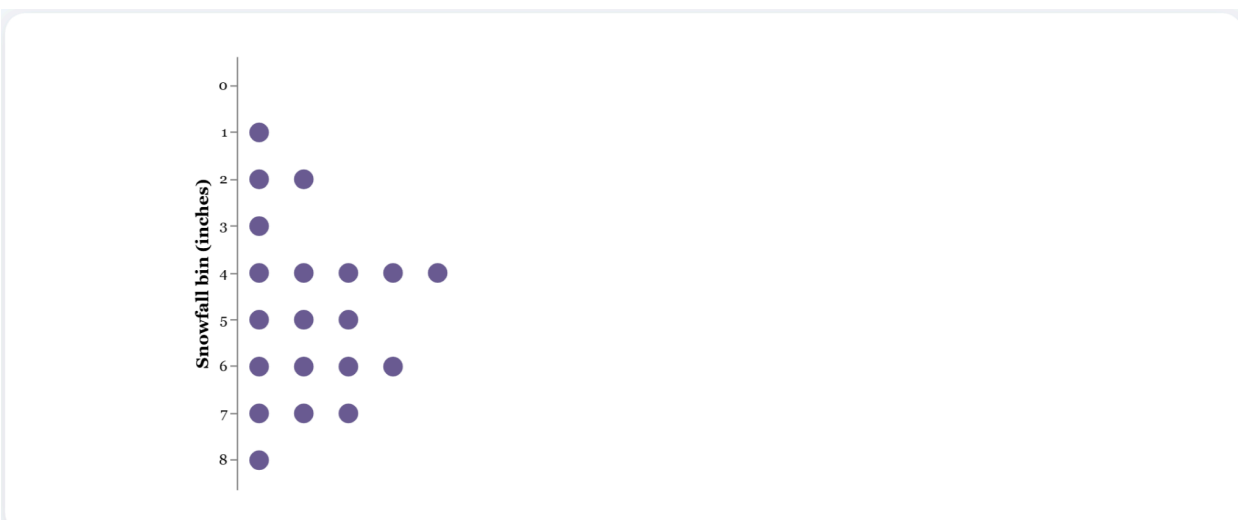

Figure 7: Vertical Quantile Dot Plot

(3) To further aid comparison between the visualization types, we next added the option for users to switch back and forth between the quantile dot plot and the bar chart–both visualizing the same information. As a fun design achievement, we made the dots of the quantile dot plots into snowflakes.

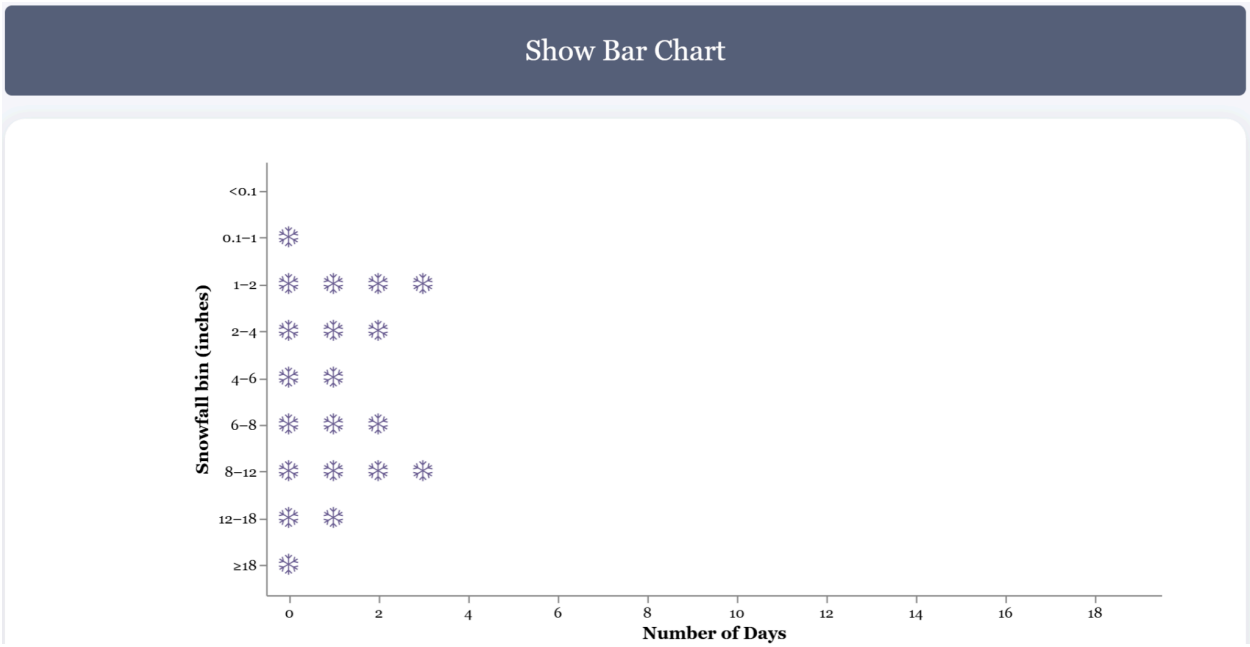

Figure 8: Visualization of Button to Toggle Between Bar Chart and Quantile Dot Plot

Implementation:

Our website "How Much Snow Might There Be?" features an interactive quantile dot plot and bar chart visualizing the likelihood of a variety of snowfall levels. By clicking "Show Dot Plot" or "Show Bar Chart", users can flip between the visualizations.

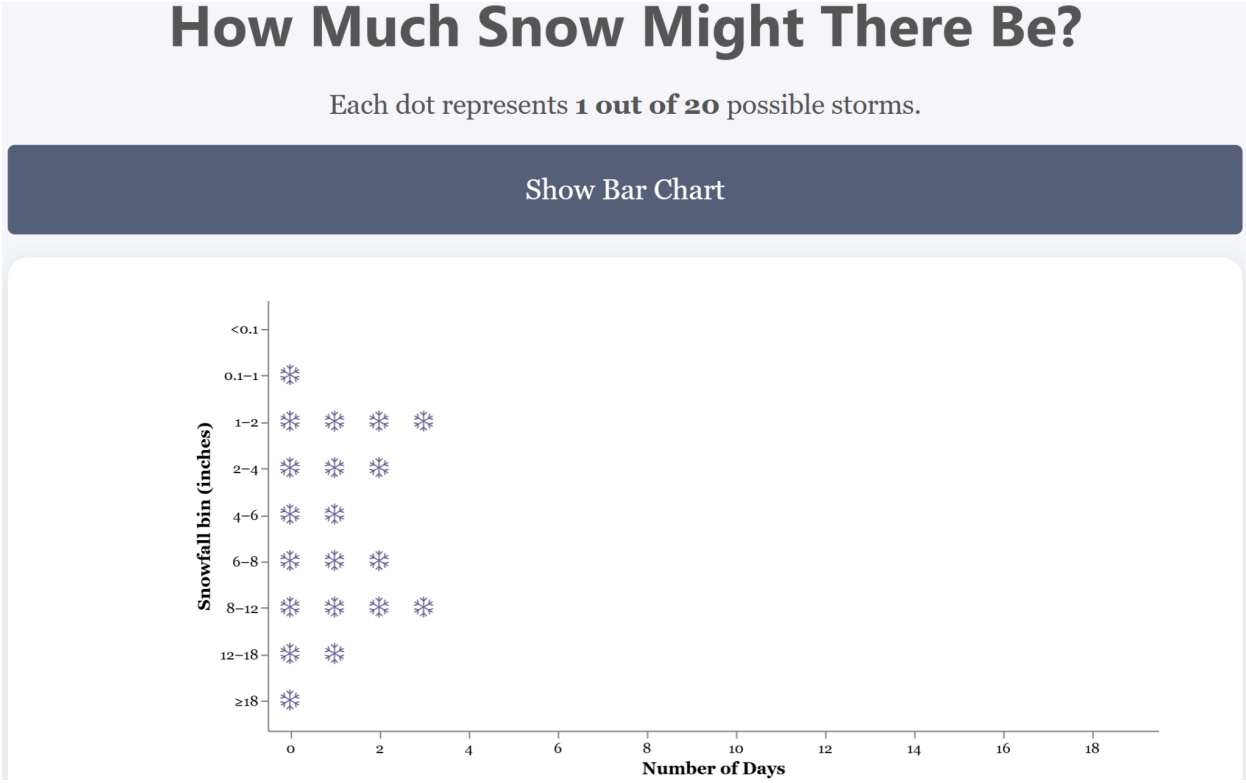

Figure 9: Interactive Button Enabling Comparison of Bar Chart and Quantile Plot Information

Users have the option to load the example dataset illustrating February 20-23, 2026 for Worcester, Massachusetts 01602 or their own. Instead of manual boxes for data inputting, we switched to sliders and required the likelihoods to equal 100%. The website communicates the total percentage users have allocated and how much they have remaining.

Figure 10: Sliders for Snowfall Likelihood and User Allocation Communication

If they try to generate visualizations with values that do not sum to 100%, the website will display "Please allocate exactly 100%. You are at [their value]%." At the bottom of the sliders, the specific dot mapping for the inputted values of the quantile dot plot is explained.

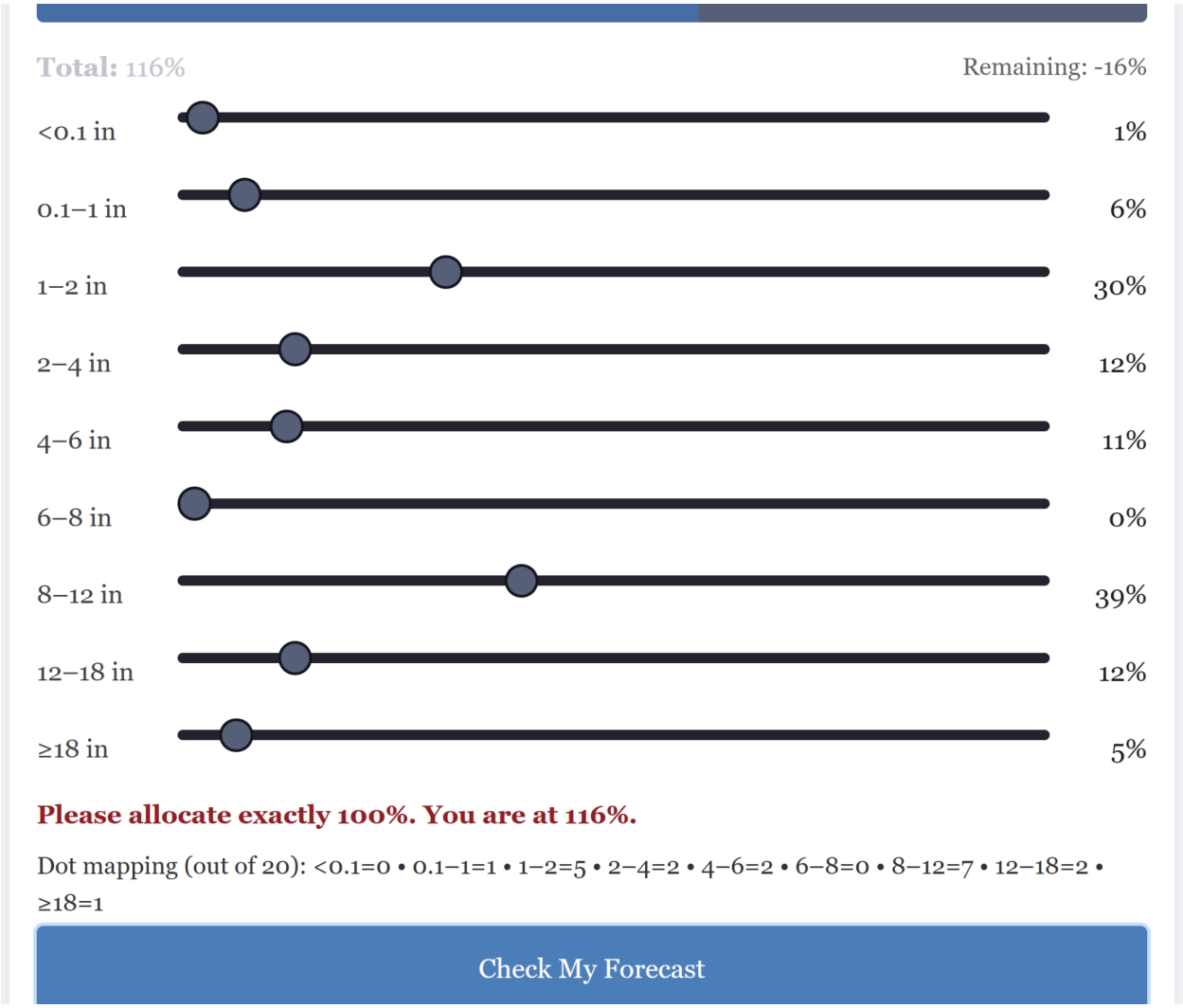

Figure 11: User Allocation Error Communication and Dot Plot Mapping

Viewers can also hover over a dot/snowflake to see the snowfall level range, likelihood percentage, and estimated days with this amount of snowfall.

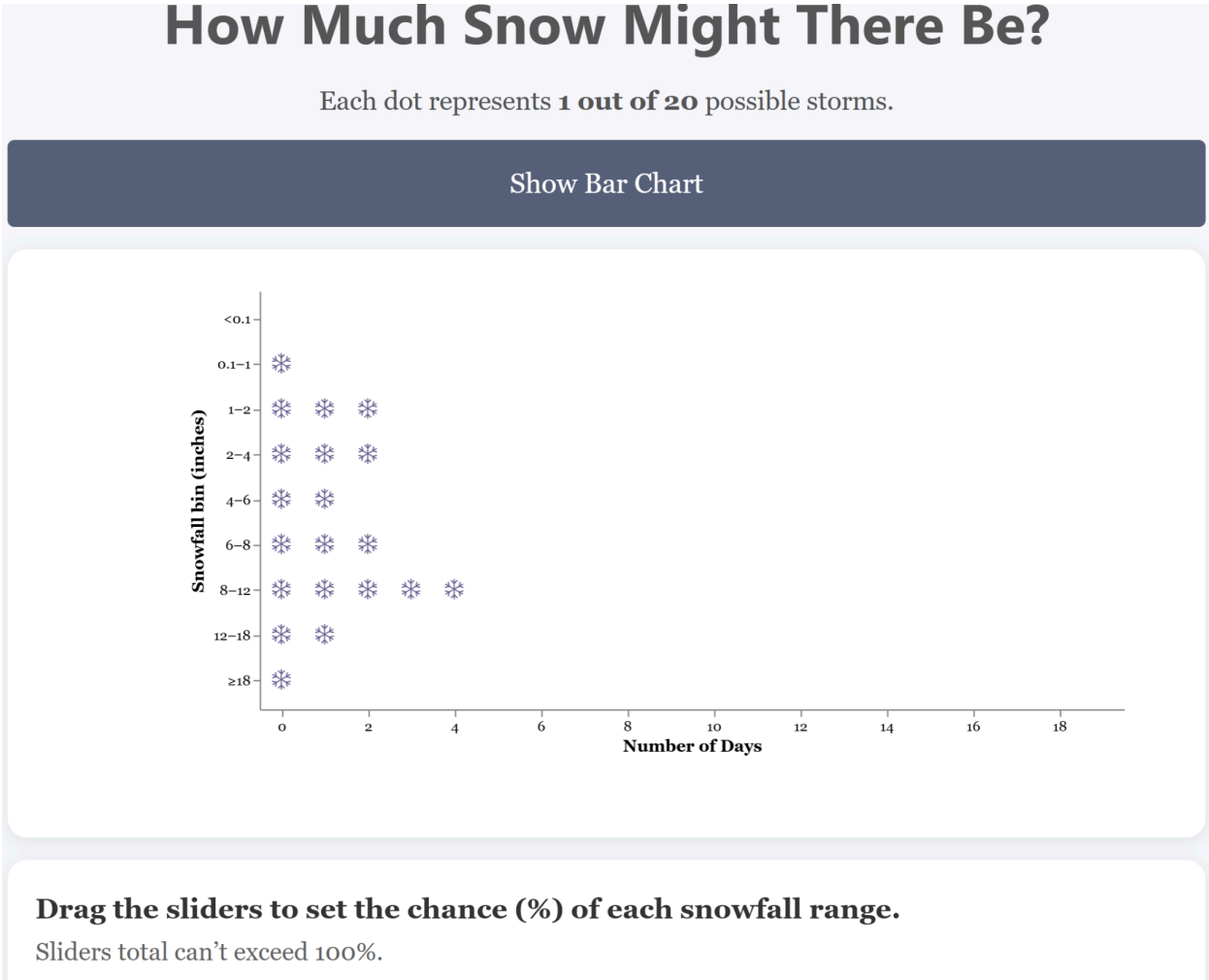

Figure 12: Example of Hovering Over a Dot in the Quantile Dot Plot

Hovering over a bar in the bar chart displays the snowfall level range and likelihood percentage.

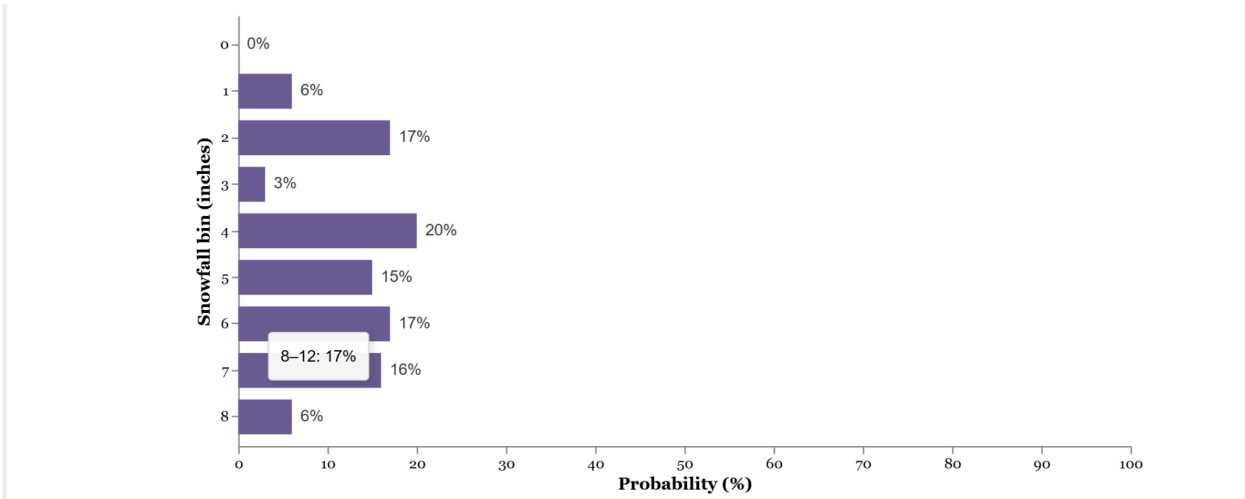

Figure 13: Example of Hovering Over a Bar in the Bar Chart

Evaluation:

We learned a more accurate interpretation of the data using the quantile dot plot instead of the bar chart. While a long list of likelihood percentages is hard to understand, visualizing the number of hypothetical days a certain snowfall level can be expected is more illuminating and interpretable.

Through literature review of a 2016 and 1987 study on the accuracy of estimates given different visualizations, we answered our question "Are quantile dot plots an effective visualization method for snowfall prediction?" with a yes. By interactively switching back and forth between a quantile dot plot and bar chart both visualizing inputted snow data and considering our takeaway from each figure, we answered our question "Do quantile dot plots help people better understand probabilistic snowfall forecasts compared to traditional bar charts?" with a yes as well. The answer to our question we conceived during the project's progression "Does a quantile dot plot being vertical as opposed to horizontal affect interpretability? Positively or negatively?" is a third yes, and positively. As we are comparing to a bar chart with horizontal data points, a quantile dot plot also with horizontal data points increases the ease of comparison.

Our visualization works well to illustrate snowfall predictions, with easy viewing on both mobile devices and computers, and we could further improve it by adding a weather API for the user to select their location and having the website input their data automatically.

Day-by-Day Process:

Day 1 Synopsis:

We suggested our project to the professor and were approved. We will use quantile dot plots to represent snowfall chances instead of the bar charts with percentages. The interactivity is a user can input the chances of each amount of inches of snowfall based on what the weather website's information says for that specific snowstorm. Our primary research question is "Do quantile dot plots help people better understand probabilistic snowfall forecasts compared to traditional bar charts?"

Day 2 Synopsis:

Each dot on the quantile dot plot represents one hypothetical day, with a predicted level of snowfall. Instead of bar chart percentages, it will be represented like out of 20 possible days, how many would have <0.1, .1-1, 1-2, 2-4, 4-6, 6-8, 8-12, 12-18, or >=18 inches. Potentially as a fun design achievement, we could see if we are able to make the dots into simple snowfalls, but that may very well not be possible. Another level of interactivity

beyond inputting values could be letting users select how many hypothetical days, like 20, 50, 100 or a slider. Based on the 2016 paper, however, a dot plot with 100 instances is less interpretable than one with 20, so maybe we should actually fix the number of days as 20. Yesterday's Vega link has some code we can consult, as well as the GitHub link from the 2016 study. Both could guide us in designing our visualization.

Day 3 Synopsis:

We clarified whether we are designing the website such that the user inputs the percentages and the quantiles and then compares how different they are, or just the dot plot. The answer is that the user inputs the percentages for each level of snowfall and then is shown a quantile dot plot.

Day 4 Synopsis:

Victor and Karinne flushed out actionable steps and visualized an outline for the website (drawn above). Victor has done some initial web development, and Aditya can add to it. Karinne found and requested the true snowfall data for the range illustrated in the example, February 20-23, 2026 in Worcester, Massachusetts 01602. The process book turned out to be a bit more involved than we were thinking given the professor's examples—including day-to-day meeting notes, problems encountered, etc—so Karinne will start working in-depth on that and link it in Slack. She will type up the motivation (why quantile plots are better than bar charts) using the 2016 paper and other studies to add to the website. One thing that might be useful is letting people toggle between the traditional bar chart and the quantile dot plot, so they can directly compare how the same probabilities are represented. That could help reinforce the research question about whether dot plots are easier to interpret. Aditya believes horizontal rows for each snowfall bin will probably read the cleanest. The team could do something like 20 dots total (each dot = 1 out of 20 possible days), so the frequency interpretation is really clear. We might also want a small indicator that shows if the percentages sum to 100% when the user enters them, just to avoid weird case issues. For the percentage error catcher, we could also communicate the invalid total like "102%" to help people know they need to remove 2%. This could be useful since there are a lot of snowfall levels, so it is hard to add everything up quickly in your head. Aditya added the button at the top to switch between bar and dot chart, as well as sliders for the likelihood percentages. The sliders must total 100%, and the website is cleaner. The website can be viewed well on both mobile devices and computers.

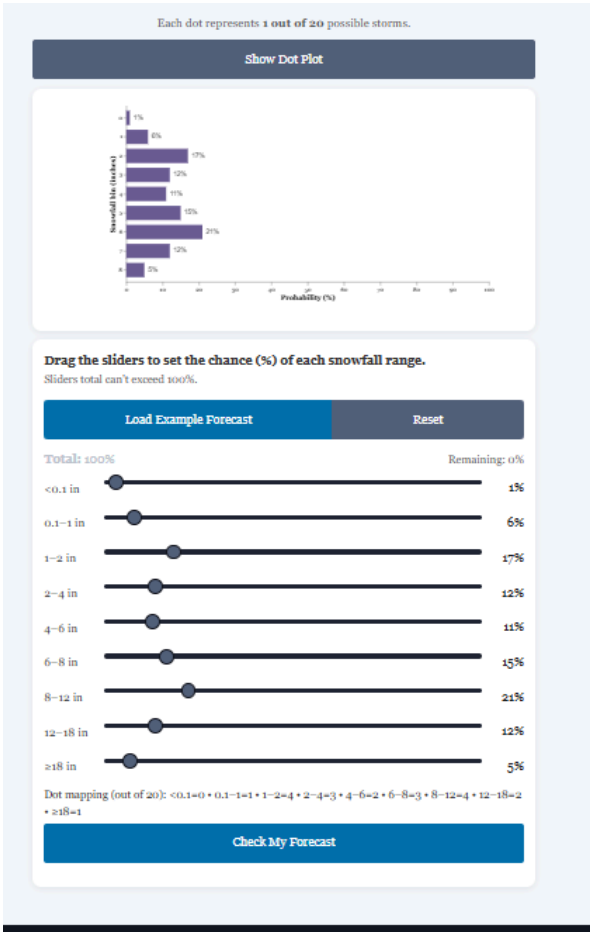

Figure 14: Mobile View of Snowfall Prediction Website Showing Bar Chart
Figure 15: Mobile View of Snowfall Prediction Website Showing Quantile Dot Plot

Day 5 Synopsis:

A first draft of the process book is complete. We have identified some potential tweaks, including changing the font color of the website title to be more readable, adding the label "Number of Days" to the quantile dot plot's x-axis, fixing that the forecast visuals still generate even if the values do not equal 100%, and adjusting that the example forecast quantile dot plot shows 4 dots for bin 2 but represents 3 of 20 days and 3 dots for bin 3 but represents 2 of 20 days.

Figure 16: Website View on Day 5

Day 6 Synopsis:

We adjusted the font color of the website title to be more readable and added the label "Number of Days" to the quantile dot plot's x-axis. The visual glitch with the example forecast still eludes us, and the interactivity of the website makes it difficult to disallow

visualization of likelihoods that do not perfectly sum to 100%. As a fun design achievement, we made the dots of the quantile dot plots into snowflakes.

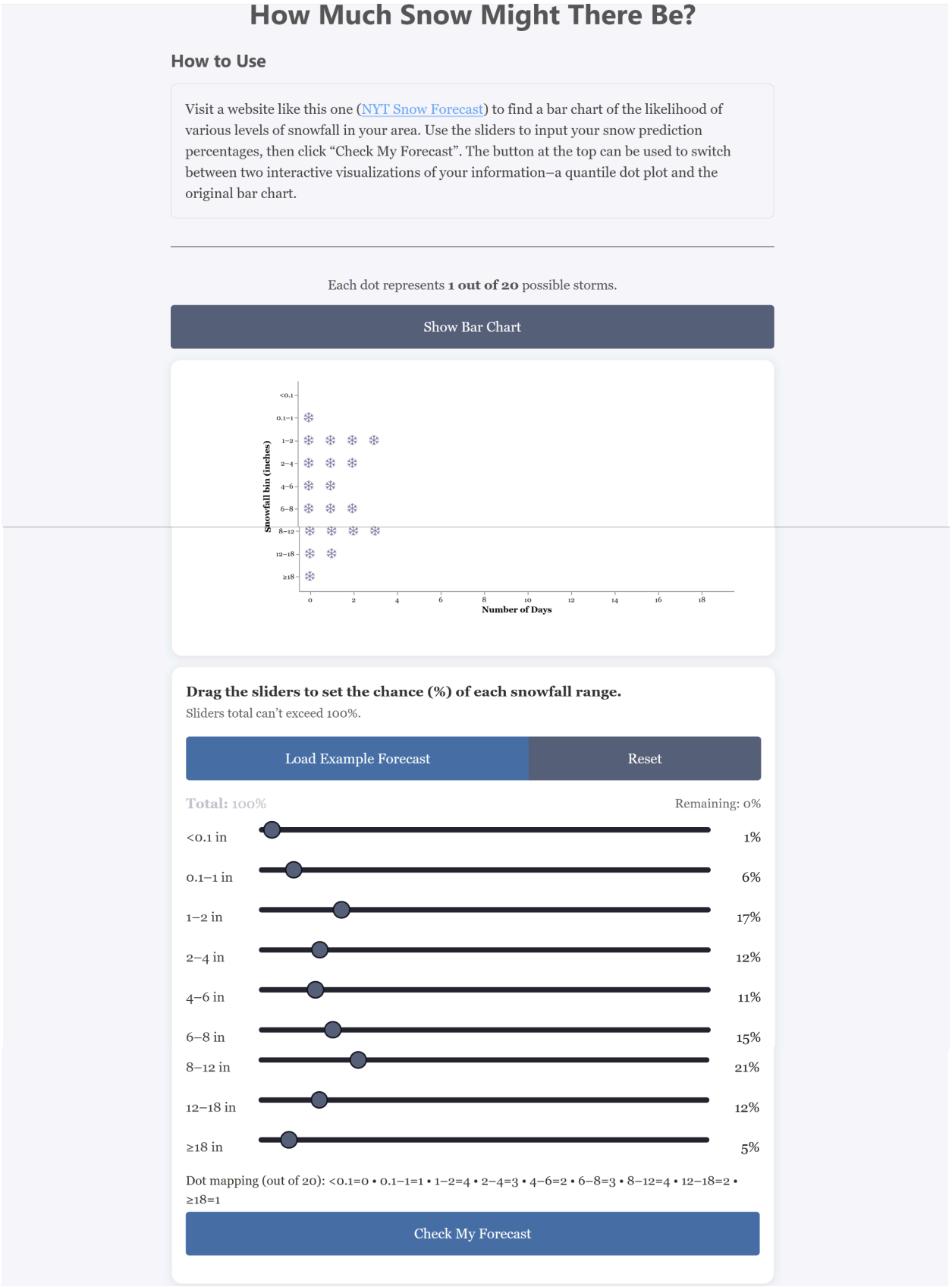

Figure 17: Website View on Day 6

References:

Bhatia, A., Katz, J., Sanger-Katz, M., & Jacoby, S. (2026, March 5). How Much Snow Will Fall Where You Live? *The Upshot Newsletter*. https://www.nytimes.com/interactive/2026/upshot/snow-forecast-totals-lookup.html

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Quantile Dot Plot Example. (n.d.). Vega. Retrieved March 5, 2026, from https://vega.github.io/vega/examples/quantile-dot-plot/